NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

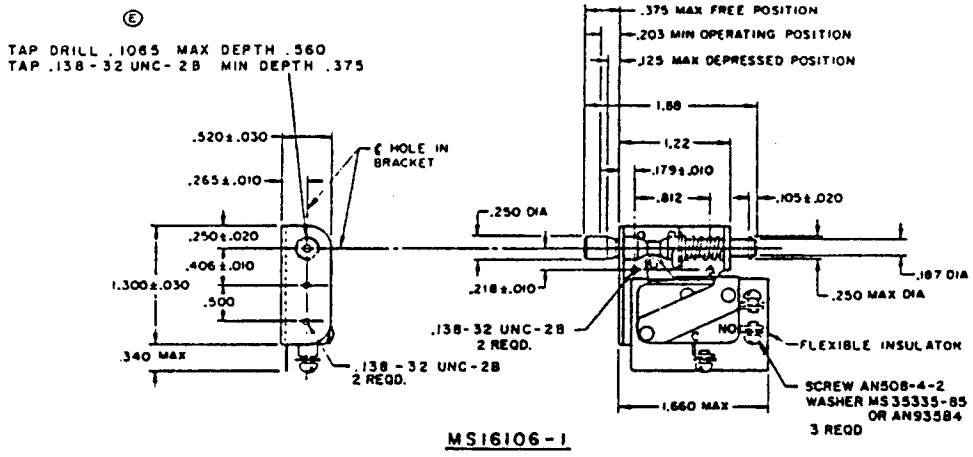
MIL-PRF-8805/56E
19 February 1976
SUPERSEDING
MIL-S-8805/56D
23 July 1973

PERFORMANCE SPECIFICATION SHEET

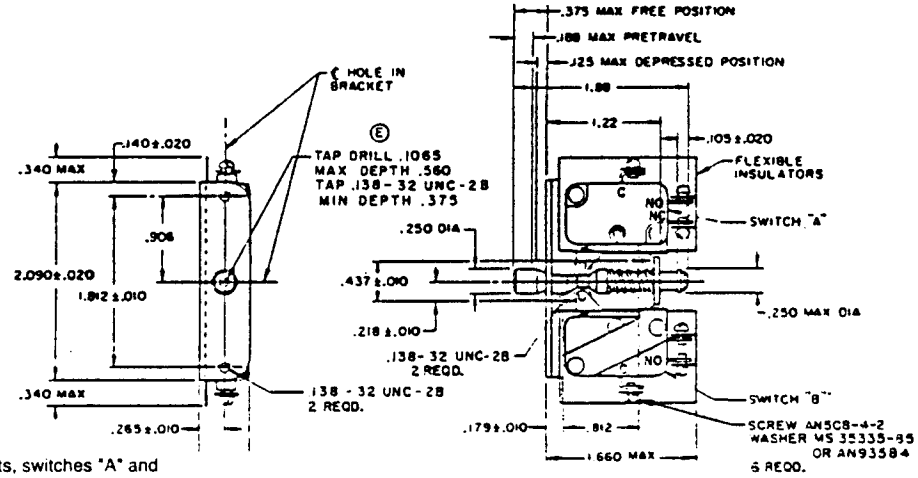
SWITCH ASSEMBLIES, SENSITIVE, INTERLOCK, UNSEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch assemblies described herein shall consist of this document and the latest issue of Specification MIL-S-8805.



MS16106-1



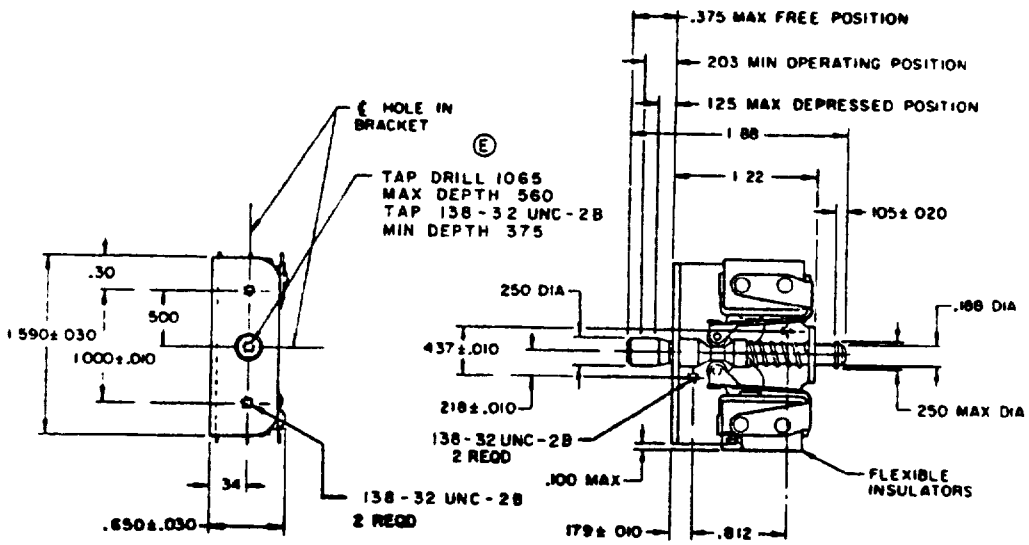
MS16106-2

On MS16106-2 units, switches "A" and "B" operate on pull stroke of actuator rod. Switch "B" operates on push stroke of actuator rod.

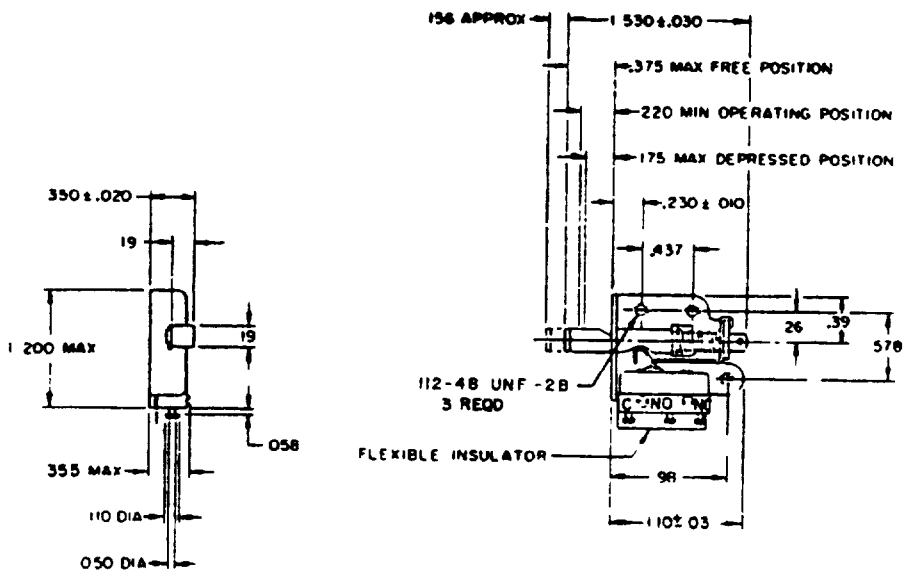
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FSC 5930

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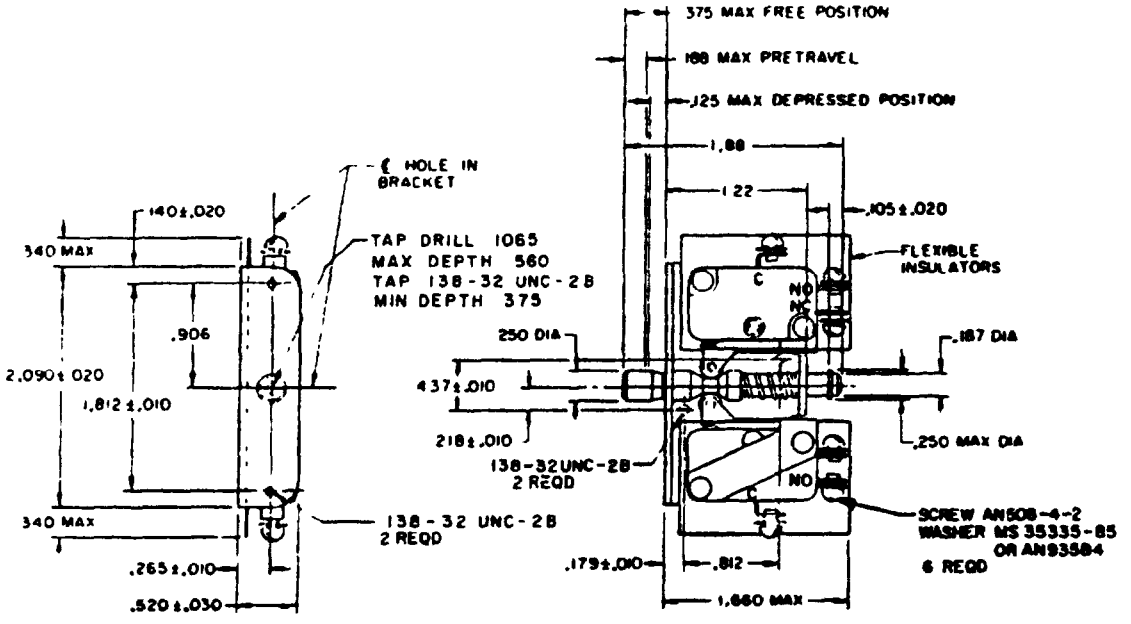


MS16106-3



MS16106-4

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MS16106-5

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.010	.25	.179	4.55	.350	8.89	1.000	25.40
.020	.51	.187	4.75	.355	9.02	1.10	27.90
.030	.76	.188	4.78	.375	9.53	1.200	30.48
.050	1.27	.19	4.83	.39	9.91	1.22	30.99
.058	1.47	.203	5.16	.406	10.31	1.300	33.02
.100	2.54	.218	5.54	.437	11.10	1.530	38.86
.105	2.67	.220	5.59	.500	12.70	1.590	40.39
.110	2.79	.230	5.84	.520	13.21	1.660	42.16
.125	3.18	.250	6.35	.578	14.68	1.812	46.02
.140	3.56	.26	6.60	.650	16.51	1.88	47.75
.156	3.96	.265	6.73	.812	20.62	2.090	53.09
.175	4.45	.30	7.62	.906	23.01		
		.340	8.64	.98	24.89		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.02 (.51 mm) for two place decimals.
4. Notch in actuator rod will permit up to .038 (.97 mm) displacement between the rod and the hole in the bracket in the free position on MS16106-1, MS16106-2, MS16106-3 and MS16106-5 and up to .062 (1.57 mm) displacement between the rod and the hole in the bracket when the rod is fully extended on MS16106-4.
5. Exact shape of switch assembly optional provided dimensions are not exceeded. Internal configuration optional.
6. Positive overtravel stop shall be provided on the rod for operation in either direction.
7. Push to operate – Returns automatically to position shown. Pull to operate – Remains in operated position until reset for automatic return by next full stroke 'push' operation.
8. All switches shall be permanently fixed to the assembly.

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REQUIREMENTS.

Dimensions and configuration: See figure
Switch assembly: Basic switch shall be as specified in table I and listed on QPL-8805
Enclosure design: 1 (Unsealed)
Temperature characteristic: 1 (-55° to +85°C)
Shock type M (100G, test condition I, method 213 of MIL-STD-202)
Vibration grade: 1 (10 to 500 Hz).
Weight: See table I.
Contact arrangement: See table I.
Operating characteristics: See table I.

TABLE I. Part number, operating characteristics, contact arrangement, and weight.

Part number MS16106-	Total travel 1/		Contact arrangement	Weight	Basic switch
	Push direction	Pull direction			
	Inch (approx)	Inch (approx)		lb max	
1	.250	.187	SPDT	09	MS25253-1
2	.250	.187	2/	14	MS25253-1
3	.250	188	4PDT	11	M8805/22-001
4	156	156	SPDT	03	M8805/22-001
5	250	187	DPDT	14	MS25253-1

1/ For information only.
2/ Switch "B" only operates on push stroke (SPDT), switches "A" and "B" operate on pull stroke (DPDT).

Contact resistance: Not applicable.
Dielectric withstanding voltage:
Sea level: 1,000 Vrms.
Altitude 1/ 400 Vrms at 50,000 ft.
In qualification inspection table after electrical endurance the dielectric withstanding voltage points of application between all unconnected terminals of the same pole is not applicable
Mechanical endurance: 25,000 cycles.
Electrical endurance. 25,000 cycles
Electrical ratings: See table II.

TABLE II. Electrical ratings.

Load	MS16106-1, -2, -5				MS16106-3, -4		
	28 Vdc	125 Vdc 1/	250 Vdc 1/	115 Vac 60 Hz	28 Vdc	115 Vac 60 Hz	250 Vac 1/ 60 Hz
	amperes	amperes	amperes	amperes	amperes	amperes	amperes
Sea level:							
Resistive	10	.5	.25	10	5	5	5
Inductive	10	.1	.1	10	3	5	5
50,000 ft:							
Resistive	10	---	---	10	5	---	---
Inductive	6	---	---	10	2 5	---	---

1/ Application information only.

Qualification (group submission): See table III.

1/ To be performed only after electrical endurance.

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TABLE III. Qualification inspection (group submission)

Examination or test	Basic switch assemblies	Additional sample units for combined submission	Extent of approval
Group I			
Visual and mechanical examination Operating characteristics	MS16106-3 and MS16106-5 (12 sample units each)	MS16106-1 (2 sample units) MS16106-2 (2 sample units) MS16106-4 (8 sample units)	All
Group II			
Strength of actuating means 1/ Strength of mounting 1/ Thermal shock Vibration Shock Operating characteristics Dielectric withstanding voltage Visual and mechanical examination	(4 sample units each from group I)	MS16106-4 (4 sample units from group I)	
Group III			
Salt spray (corrosion) Visual and mechanical examination	(2 sample units each from group I)	---	
Group IV			
Low temperature operation (2 sample units only) Mechanical endurance at low temperature (2 units) 2/ Mechanical endurance at high temperature (2 units) Operating characteristics Dielectric withstanding voltage Visual and mechanical examination	(4 sample units each from group I	MS16106-4 (4 sample units from group I)	
Group V			
Electrical endurance Resistive load, 115 Vac	(2 sample units each from group I)	---	

1/ Two sample units only.
2/ Same sample units as for low temperature operation.

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Army - EL
Navy - EC
Air Force - 11

Review activities:
Army -
Navy -
Air Force - 17, 80, 85
DSA - ES

User activities:
Army - MI, AV, WC
Navy - AS, MC, OS
Air Force - 19

Preparing activity
Navy - EC

Agent:
DSA - ES

(Project 5930-1009-1)

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	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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